

Name: Sagi Sahasra Srihasa
Roll : 2520030008

1)

Program:

```
sahasrasagi@Sahasra: ~/OS_L  ×  +  ▾
GNU nano 8.7.1
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], wt[20], tat[20];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time for each process:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

Output:

```
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ ./fcfs
Enter number of processes: 3
Enter burst time for each process:
P1: 5
P2: 3
P3: 8

Process BT      WT      TAT
P1      5       0       5
P2      3       5       8
P3      8       8      16
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ |
```

2)

Program:

```
GNU nano 8.7.1
#include <stdio.h>

int main() {
    int n, i, j;
    int bt[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (bt[i] > bt[j]) {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

Output:

```
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ nano sjf_non-preemptive.c
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ gcc sjf_non-preemptive.c -o sjf_non-preemptive
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ ./sjf_non-preemptive
Enter number of processes: 4
Enter burst time:
P1: 5
P2: 3
P3: 8
P4: 2

Process BT      WT      TAT
P1      2        0        2
P2      3        2        5
P3      5        5       10
P4      8       10       18
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ |
```

3)

Program:

```
GNU nano 8.7.1
#include <stdio.h>

int main() {
    int n, i, j;
    int bt[20], priority[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time and priority:\n");

    for (i = 0; i < n; i++) {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }

    // Sort according to priority
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (priority[i] > priority[j]) {
                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t\t%d\t%d\n",
            i + 1, bt[i], priority[i], wt[i], tat[i]);
    }

    return 0;
}
```

Output:

```
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ nano priority.c
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ gcc priority.c -o priority
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ ./priority
Enter number of processes: 3
Enter burst time and priority:
P1 Burst Time: 2
P1 Priority: 3
P2 Burst Time: 4
P2 Priority: 2
P3 Burst Time: 1
P3 Priority: 2

Process BT      Priority      WT      TAT
P1      4        2           0        4
P2      1        2           4        5
P3      2        3           5        7
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ |
```

4)Round Robin

Program:

```
GNU nano 8.7.1
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], rem_bt[20];
    int wt[20], tat[20];
    int quantum;
    int time = 0;
    int done;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
        rem_bt[i] = bt[i];
        wt[i] = 0;
    }

    printf("Enter time quantum: ");
    scanf("%d", &quantum);

    // Round Robin
    while (1) {
        done = 1;

        for (i = 0; i < n; i++) {
            if (rem_bt[i] > 0) {
                done = 0;

                if (rem_bt[i] > quantum) {
                    time = time + quantum;
                    rem_bt[i] = rem_bt[i] - quantum;
                }
                else {
                    time = time + rem_bt[i];
                    wt[i] = time - bt[i];
                    rem_bt[i] = 0;
                }
            }
        }

        if (done == 1)
            break;
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

Output:

```
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ nano round_robin.c
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ nano round_robin.c
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ gcc round_robin.c -o round_robin
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ ./round_robin
Enter number of processes: 3
Enter burst time:
p1: 2
p2: 3
p3: 4
Enter time quantum: 2

Process BT      WT      TAT
p1      2      0      2
p2      3      4      7
p3      4      5      9
sahasrasagi@Sahasra:~/OS_Lab/Scheduling_Algorithms$ |
```